

Experiment - 15

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Course Code: MLA0305

Slot: B

Aim

To simulate a **Multi-Agent Reinforcement Learning environment** and demonstrate a **Partially Observable Markov Decision Process (POMDP)** using multiple agents.

Algorithm / Procedure

1. Initialize two agents in the environment.
2. Set the goal position for both agents.
3. Allow each agent to independently select an action.
4. Update the position of each agent based on its action.
5. Calculate the reward based on the agents' positions.
6. Continue the simulation until the agents reach the goal.
7. Record the positions and rewards at each step.
8. Plot the movement of both agents and their rewards.

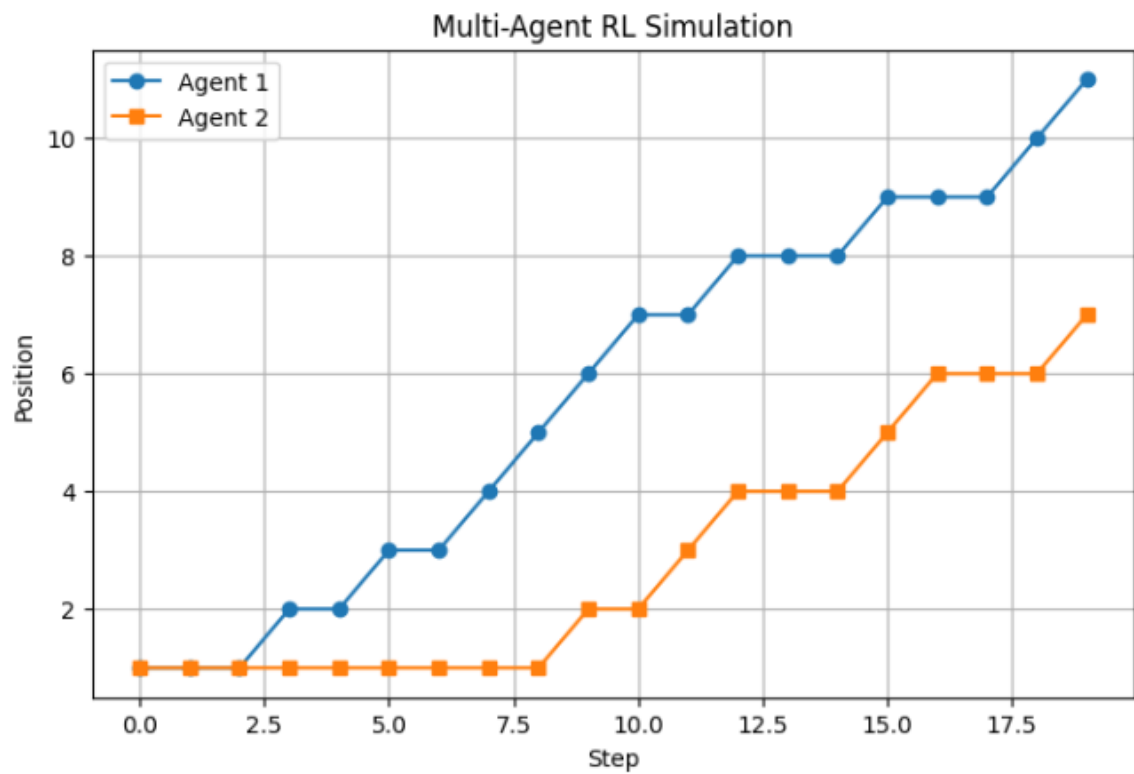
Code

```
import numpy as np
import matplotlib.pyplot as plt

class MultiAgentEnv:
    def __init__(self):
        self.positions=[0,0]
        self.goal=10
    def reset(self):
        self.positions=[0,0]
        return self.positions
    def step(self,actions):
        for i in range(2):
            if actions[i]==1:
                self.positions[i]+=1
        reward=sum(self.positions)/2
```

```
        done=all(p>=self.goal for p in self.positions)
        return self.positions,reward,done
env=MultiAgentEnv()
positions=env.reset()
agent1=[]
agent2=[]
rewards=[]
for step in range(20):
    actions=[np.random.randint(0,2),np.random.randint(0,2)]
    positions,reward,done=env.step(actions)
    agent1.append(positions[0])
    agent2.append(positions[1])
    rewards.append(reward)
    print("Step:",step+1,"Agent 1:",positions[0],"Agent 2:",positions[1],"Reward:",reward)
    if done:
        break
plt.figure(figsize=(8,5))
plt.plot(agent1,marker="o",label="Agent 1")
plt.plot(agent2,marker="s",label="Agent 2")
plt.xlabel("Step")
plt.ylabel("Position")
plt.title("Multi-Agent RL Simulation")
plt.legend()
plt.grid(True)
plt.show()
```

Output



Result

The multi-agent reinforcement learning environment was successfully simulated, and the movements and rewards of the agents were visualized using a graph.